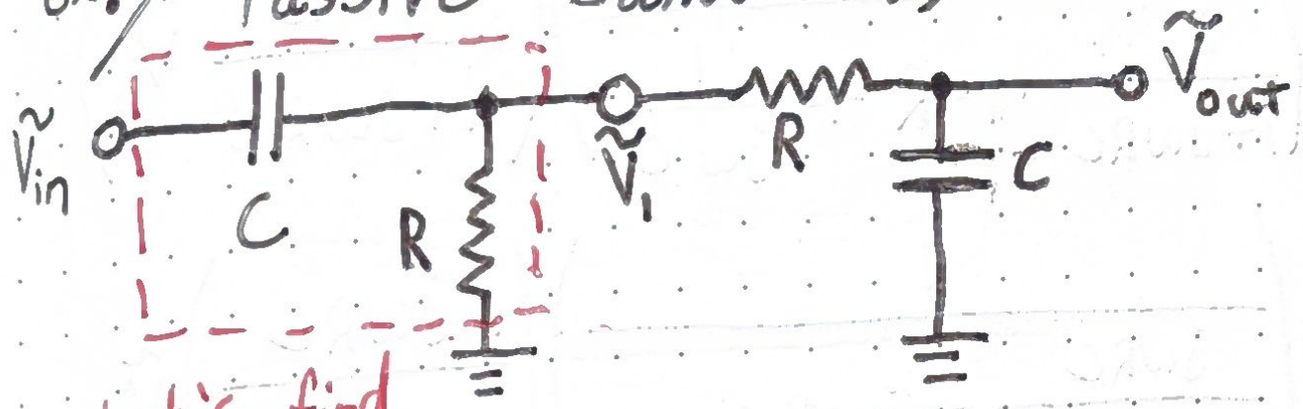
Thévenin for Combining Transfer Functions:

ex./ Passive Band-Pass



Let's find the Thévenin of the HP Stage

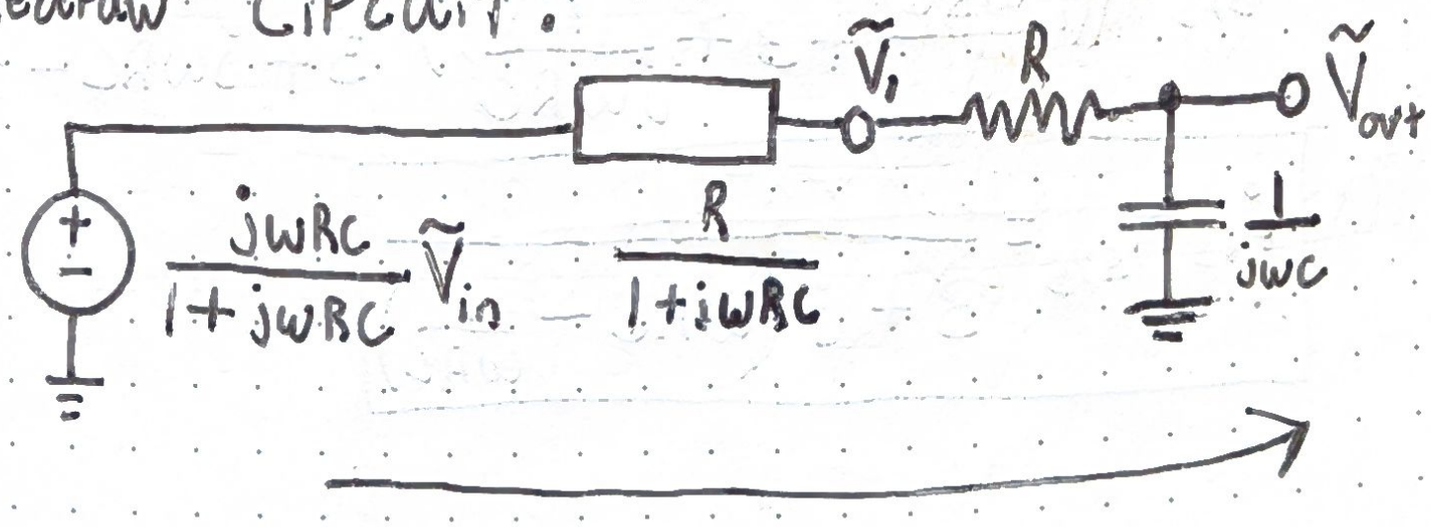
$$\tilde{V}_{oc} = \frac{R}{R + 1/j\omega C}\tilde{V}_{in} = \frac{j\omega RC}{1 + j\omega RC}\tilde{V}_{in}$$

$$\tilde{I}_{sc} = \frac{\tilde{V}_{in}}{1/j\omega C} = j\omega C\tilde{V}_{in}$$

$$\Rightarrow Z_{Th} = \frac{\tilde{V}_{oc}}{\tilde{I}_{sc}} = \left(\frac{\cancel{j\omega RC}}{1 + j\omega RC}\cancel{\tilde{V}_{in}}\right)\left(\frac{1}{\cancel{j\omega C}\,\cancel{\tilde{V}_{in}}}\right)$$

$$\Rightarrow Z_{Th} = \frac{R}{1 + j\omega RC}$$

Redraw Circuit:



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Now Use Volt division to find H_{SYS}

$$\tilde{V}_{out} = \left(\frac{1/j\omega C}{\frac{R}{1+j\omega RC} + R + \frac{1}{j\omega C}} \right) \left(\frac{j\omega RC}{1+j\omega RC} \tilde{V}_{in} \right)$$

$$\Rightarrow \tilde{V}_{out} = \left(\frac{1}{\frac{j\omega RC}{1+j\omega RC} + j\omega RC + 1} \right) \left(\frac{j\omega RC}{1+j\omega RC} \right) \tilde{V}_{in}$$

$$\Rightarrow \tilde{V}_{out} = \left(\frac{1+j\omega RC}{j\omega RC + (1+j\omega RC)^2} \right) \left(\frac{j\omega RC}{1+j\omega RC} \right) \tilde{V}_{in}$$

let $j\omega RC = X$ and $X^2 = -\omega^2 R^2 C^2$

$$\Rightarrow \frac{\tilde{V}_{out}}{\tilde{V}_{in}} = H_{SYS} = \left(\frac{\cancel{1+X}}{X + (1+X)^2} \right) \left(\frac{X}{\cancel{1+X}} \right)$$

$$\Rightarrow \frac{\tilde{V}_{out}}{\tilde{V}_{in}} = H_{SYS} = \frac{X}{X^2 + 3X + 1} = \frac{j\omega RC}{-\omega^2 R^2 C^2 + 3j\omega RC + 1}$$

$$\Rightarrow H_{SYS} = \frac{1}{-\frac{\omega RC}{j} + 3 + \frac{1}{j\omega RC}} = \frac{1}{3 + j\omega RC - \frac{j}{\omega RC}}$$

$$\boxed{\therefore H_{SYS} = \frac{1}{3 + j\left(\omega RC - \frac{1}{\omega RC}\right)}}$$